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ACADEMY

Video 5 · Mensuration, Trig & Vectors Worked Solutions

Cambridge IGCSE Mathematics 0580, Paper 2 Non-calculator (Extended). Full worked solutions to Questions 20–23 of the NurNabi Academy Full Revision Paper, solved the simplest way, with the method and accuracy marks shown at every step, the way a Cambridge examiner awards them.

4	20	Q20–Q23	20:50
questions	marks	E5.3 – E7.4	video runtime

M1 method mark A1 accuracy mark (needs the M mark) B1 independent mark

Question 20

Sectors and cones

Sector: radius 12 cm, angle 150°. Fraction of the circle = $\frac{150}{360} = \frac{5}{12}$

(a) Find the arc length *AB*, in terms of π. [2]

Arc = $\frac{150}{360} \times 2\pi r = \frac{5}{12} \times 2 \times \pi \times 12$ M1

$= \frac{5}{12} \times 24\pi = 5 \times 2\pi$

10π cm A1

(b) Find the area of the sector, in terms of π. [2]

Area = $\frac{150}{360} \times \pi r^2 = \frac{5}{12} \times \pi \times 144$ M1

$= 5 \times 12\pi$

60π cm² A1

Tip: simplify $\frac{150}{360}$ to $\frac{5}{12}$ first. Since 12 goes into 24 and 144, no calculator is needed.

Question 20 (continued)

Sectors and cones

(c) The sector is folded into a hollow cone (*OA* meets *OB*). Find its volume in terms of π, in simplest surd form. [3]

1. Base radius: the arc becomes the base circumference:

$2\pi r = 10\pi \rightarrow r = 5$ **M1**

2. Height: the slant height is the old radius, 12. Pythagoras:

$h = \sqrt{(12^2 - 5^2)} = \sqrt{(144 - 25)} = \sqrt{119}$ **M1**

3. Volume $= \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 25 \times \sqrt{119}$

$\frac{25\sqrt{119}}{3}\pi$ cm³ **A1**

Key idea: when you fold a sector into a cone
- arc length → base circumference
- radius of sector → **slant height** *l*

√119 is already simplest: 119 = 7 × 17, with no square factor.

Common mistake: using 12 as the height. 12 is the slant edge, not the vertical height.

Question 21

Exact trigonometric values and trig equations

(a) Write down the exact value of cos 30°. [1]

$\frac{\sqrt{3}}{2}$ **B1**

Learn this table:

	30°	45°	60°
sin	1/2	√2/2	√3/2
cos	√3/2	√2/2	1/2
tan	√3/3	1	√3

(b) Triangle PQR: angle $Q = 90°$, angle $P = 30°$, $PQ = 6$. Find the exact value of x (= QR). [2]

QR is **opposite** 30°, and PQ is **adjacent**, so use tan:

$\tan 30° = \frac{x}{6} \rightarrow x = 6 \tan 30° = 6 \times \frac{\sqrt{3}}{3}$ **M1**

$x = 2\sqrt{3}$ **A1**

If you use $\tan 30° = \frac{1}{\sqrt{3}}$, you get $\frac{6}{\sqrt{3}}$. Rationalise: $\frac{6\sqrt{3}}{3} = 2\sqrt{3}$.

Question 21 (continued)

Exact trigonometric values and trig equations

(c) Solve $2\sin x + \sqrt{3} = 0$ for $0° \le x \le 360°$. [3]

Rearrange: $\sin x = -\frac{\sqrt{3}}{2}$ B1

$\sin 60° = \frac{\sqrt{3}}{2}$, so the **reference angle** is 60°.

sin is **negative** in the 3rd and 4th quadrants:

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3rd quadrant: 180 + 60 = 240
4th quadrant: 360 - 60 = 300
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$x = 240°$ or $x = 300°$ B1 B1

CAST diagram (anticlockwise from the right): **A**ll positive (0–90°), **S**in positive (90–180°), **T**an positive (180–270°), **C**os positive (270–360°). Sin is negative in T and C.

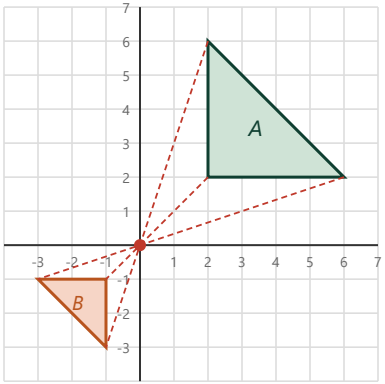
Common mistake: giving −60°, which is outside the range, or 120°, where sin is positive.

Sketch check: the sine curve is below the axis between 180° and 360°, so both answers are there ✓

Question 22

Transformations — negative scale factor

Describe fully the **single** transformation that maps triangle *A* onto triangle *B*. [3]



1. Type: *B* is smaller and **upside down on the opposite side of the origin**, so it's an enlargement with a **negative** scale factor.

2. Scale factor: *A*'s side is 4 squares and *B*'s is 2 squares, so size × ½. It is inverted, so −½.

3. Centre: join matching corners. The lines cross at (0, 0).

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(2, 2) → (−1, −1)
(2, 6) → (−1, −3)
(6, 2) → (−3, −1)   each = -1/2 × A
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Enlargement, scale factor −½, centre (0, 0) B1 B1 B1

Three facts, one mark each: the word *enlargement*, the scale factor, the centre. Describing **two** transformations (e.g. rotation + enlargement) scores 0.

Question 23

Vector geometry

OA = **a**, **OB** = **b**, $AP : PB = 1 : 3$

(a) Find AB in terms of a and b. [1]

Route from A to B through O: **AB** = **AO** + **OB** = −**a** + **b**

b − a B1

(b) Find OP in its simplest form. [2]

$AP : PB = 1 : 3$, so P is $\frac{1}{4}$ of the way from A to B.

OP = **OA** + **AP** = **a** + $\frac{1}{4}$(**b** − **a**) M1

= **a** − $\frac{1}{4}$**a** + $\frac{1}{4}$**b**

$\frac{3}{4}$**a** + $\frac{1}{4}$**b** A1

Ratio trap: 1 : 3 has 4 parts, so it's $\frac{1}{4}$, not $\frac{1}{3}$.

Question 23 (continued)

Vector geometry

(c) $\mathbf{OQ} = 3\mathbf{a} + \mathbf{b}$. Explain why *O*, *P* and *Q* lie on a straight line. [1]

Compare with $\mathbf{OP} = \frac{3}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}$:

$\mathbf{OQ} = 3\mathbf{a} + \mathbf{b} = 4(\frac{3}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}) = 4\mathbf{OP}$

$\mathbf{OQ} = 4\mathbf{OP}$, so parallel with a common point *O* B1

For the mark you need both facts:
1. One vector is a **multiple** of the other (so they're parallel)
2. They **share a point** (both start at *O*)

Common mistake: only saying "they are parallel". Parallel lines can be side by side, so the shared point *O* is what makes it one straight line.

Answers at a glance

Video 5 · Mensuration, Trig & Vectors · 20 marks

Q	Topic	Answers	Marks
20	Sectors and cones	(a) 10π (b) 60π (c) $\frac{25\sqrt{119}}{3}\pi$	7
21	Exact trig values, trig equations	(a) $\frac{\sqrt{3}}{2}$ (b) $2\sqrt{3}$ (c) 240°, 300°	6
22	Negative enlargement	Enlargement, s.f. $-\frac{1}{2}$, centre (0, 0)	3
23	Vector geometry	(a) $\mathbf{b} - \mathbf{a}$ (b) $\frac{3}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}$ (c) $\mathbf{OQ} = 4\mathbf{OP}$	4
	Total		**20**

Mark allocations follow the style of the Cambridge 0580 mark scheme. The questions are original NurNabi Academy practice questions, not a prediction of the real paper. Next: **Video 6 · Probability and statistics (Q24–Q26)**.